Deep Patel

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PROFESSIONAL SUMMARY

High school senior focused on aerospace engineering and rocket propulsion. Creator of ORBIX, an aerospace engineering web app, and designer of a 3D-printed rocket bell nozzle studied with CFD simulation. Dedicated youth leader and volunteer looking for opportunities to learn more about physics, engineering, and space science through hands-on projects and research.

EDUCATION

Chelmsford High School, Chelmsford, MA Expected Graduation: June, 2027

GPA: 4.01 (Weighted)

Relevant Courses: AP Physics C, AP Computer Science, Calculus, Advanced Engineering Principles & 3D Design

SKILLS AND INTERESTS

Technical Skills: Java, CAD (Fusion 360, Revit, Onshape, OpenSCAD), CFD Simulation (SimFlow), Robotics, Microsoft Office Suite, Mathematical Modeling, Adobe Creative Cloud

Interests: Aerospace Engineering, Mechanical Engineering, Astrophysics, Rocketry, 3D Modeling

TECHNICAL PROJECTS

Aerodynamic Bell Nozzle Sep. 2025 - Present

Lead Designer & Fabricator (Self-Directed)

- Studied nozzle thermodynamics and isentropic flow independently to size a LOX/RP-1 bell nozzle for a 100 bar chamber expanding to 0.7 bar.
- Modeled the nozzle contour in OpenSCAD and checked its 15.92 area ratio against the Mach 3.6 isentropic design point.
- Set up and debugged a compressible-flow CFD case in SimFlow, tracing solver divergence through mesh, boundary condition, and limiter changes.
- 3D-printed the 84 mm nozzle in PLA, printing it inverted so no support material formed inside the bell.

ORBIX Web Application Aug. 2026 - Present

Creator & Project Lead | https://orbix-aerospace.vercel.app/

- Conceived and designed a web app with 13 engineering calculators, a mission planner, and profiles of 10 real aircraft and launch vehicles.
- Wrote the specifications for each tool, including equations, inputs, and units, and directed the programming of the front-end logic and physics engines using AI coding tools.
- Planned a verification page that checks results against published references (NACA Report 1135, U.S. Standard Atmosphere 1976); 26 of 36 values match within rounding.

ACADEMICS AND EXTRACURRICULARS

Track and Field Team Nov. 2023 - Present

Captain & Athlete

- 3x MSTCA Div. II State Relay Medalist (High Jump State Relay: 4th, 4th, 6th) and 6-season varsity athlete (Indoor & Outdoor).
- Compete in high jump, 55m hurdles, and 110m hurdles. Lead daily workouts for 75+ athletes and mentor 35+ underclassmen on form and discipline.

Artificial Intelligence (AI) Club Sep. 2025 - Present

Vice President

- Manage 15 members and host bi-weekly meetings focused on safe AI usage and coding ethics in the real world.
- Directed a project team for the MIT URTC conference, building a custom AI software tool that monitors home environments to optimize care for household plants.

Robotics Club Sep. 2023 - Jun. 2024

Member

- Worked with a team to design, build, and test robots for VEX Robotics competitions.
- Programmed the robot to navigate competition mazes autonomously and engineered the physical mechanisms needed to lift plastic balls into scoring baskets.

National Honor Society (NHS) Mar. 2026 - Present

Member & Volunteer

WORK EXPERIENCE

ALPHAPEP LLC, Lowell, MA May 2026 - Present

Email Marketing Associate

- Design and distribute targeted marketing emails to advertise a diverse variety of the company's products.
- Manage and execute outgoing email campaigns to ensure successful delivery of promotional materials and drive customer engagement.

LEADERSHIP AND VOLUNTEERING

Bachat Kumbh Project, Lowell, MA Sep. 2025 - Present

Co-Founder & Logistics Lead

- Bridged engineering and community service by designing and 3D-printing custom, temple-shaped donation boxes (Kumbhs) for local youth.
- Managed the distribution, tracking, and collection logistics of the physical boxes across the community.
- Successfully mobilized youth participation to raise $2,500+ to directly fund the construction of the new Tyngsborough temple.

BAPS Swaminarayan Sanstha (Bal Mandal & Kishore Mandal), Lowell, MA Aug. 2022 - Present

Youth Activities Coordinator & Volunteer *Aug. 2024 - Present*

Teacher *Aug. 2022 - Aug. 2024*

- Lead weekly youth assemblies for 70+ Pre-K–8th grade children and manage a team of 15 high school peer teachers.
- Plan and oversee weekly lessons covering cultural values, history, and leadership skills.
- Serve as a logistics volunteer for the high school teen wing (Kishore Mandal), helping coordinate stage setup and crowd control for major cultural festivals serving over 1,000 attendees.

BAPS Charities, Lowell, MA Jun. 2017 - Present

Walkathon Organizer & Outreach Volunteer

- Organized the annual 'In the Joy of Others' Walk-Run for 650+ participants, setting up routes and managing event registration.
- Helped raise more than $165,000 over four years for the Lowell Police Dept., Tyngsborough Fire Dept., local public schools, and The Nature Conservancy's Plant a Billion Trees campaign through the Walk Green initiative.

BAPS Akshardham, Robbinsville, NJ Jul. 2023 - Oct. 2023

Full-Time Volunteer

- Designed detailed architectural pieces using CorelDRAW and Adobe Illustrator, and created CNC machine files for a large cultural building project.
- Ran industrial CNC machines and hot wire cutters to make complex parts out of wood, PVC, and foam at scale.
- Worked with senior designers and other volunteers to ensure designs were accurate and finished on time without wasting materials.